SOC ANALYST TRIAGE TRAINING GUIDE

INVESTIGATING TEMPORAL THREATS (CERT R5.2 ENVIRONMENT)

Welcome to the interactive SOC Triage Training Portal. As an analyst evaluating the CERT r5.2 environment, your objective is to investigate system alerts, identify malicious behavior, and correctly categorize insider threat signatures. To succeed, you must approach insider threats not as isolated events, but as a timeline of escalating actions.

1. The Nature of Insider Threats: The Breadcrumb Trail

Sophisticated insider threats are rarely executed in a single, simultaneous burst. They span days, weeks, or even months. You are not looking for a single smoking gun on a single day; you are looking for a trail of breadcrumbs.

Consider how the specific threat scenarios within this environment escalate across timelines:

- **Scenario 1 (Data Exfiltration & Sabotage):** A user might begin logging in after hours, subsequently connecting a removable drive, and eventually navigating to specific external sites to upload data, triggering a high Wikileaks Access Count. These actions string together over time rather than occurring simultaneously.
- **Scenario 2 (Intellectual Property Theft):** A departing employee might spend days or weeks browsing job hunting websites or soliciting competitors. The actual exfiltration—where they use a thumb drive to steal intellectual property before leaving—happens much later in the timeline, requiring you to link the earlier job-seeking behavior to the later data theft.
- **Scenario 3 (Credential Compromise):** An attacker might download a keylogger to a thumb drive on a Tuesday. The actual exploitation—an after-hours unauthorized login to a supervisor's machine—might not happen until Friday.
- **Scenario 4 (Prolonged Exfiltration):** A compromised user logs into another user's machine, searches for sensitive files, and emails them to a personal account. This behavior starts small and increases in frequency over a rolling three-month period.

2. The Detection Engine: Understanding the AI's Limitation

The underlying detection engine monitoring this network is a Bidirectional LSTM (BiLSTM) autoencoder. It is critical to understand how this AI evaluates risk:

> **Critical System Constraint:** The BiLSTM autoencoder operates on a strict 24-hour daily resolution.

When a user's daily behavior deviates from their historical baseline, the autoencoder's reconstruction error spikes, generating an anomaly alert on your timeline. Because the AI evaluates data day-by-day, it flags distinct steps (like a keylogger download on Tuesday and an unauthorized login on Friday) as separate anomalies. It cannot automatically link them.

Your job is to investigate those days, identify the behavioral indicators, and connect the dots across the timeline to determine the full scenario.

3. Your Investigative Environments

Depending on your assigned cohort phase, you will utilize one of two dashboard interfaces to conduct your investigations. Both feature an interactive Gemini LLM assistant, but they differ in how they present the underlying data.

Design A (Traditional Baseline + LLM)

This interface utilizes standard Z-score tables and raw user action logs to identify statistical outliers. A high Z-score indicates that a behavior occurred at a mathematically unusual frequency. The integrated LLM is configured to help you interpret the autoencoder's input feature tables and clarify raw user actions, guiding your manual hunt through the statistical noise.

Design B (XAI Visual Suite + LLM)

This interface overlays the detection model with SHAP (SHapley Additive exPlanations) visual attribution plots. The LLM acts as an analytical translator for these specific visualizations.

- Use the **Local Waterfall Plot** to see the precise features driving a specific day's threat score.
- Use the **Global Beeswarm Plot** to analyze how those threat indicators aggregate over the entire three-month investigation timeline.

4. Standard Operating Procedure (SOP) for Triage

To effectively match threat indicators to specific scenarios, follow this structured workflow during your evaluation:

1. **Identify the Initial Spike:** Use the main timeline to spot when the autoencoder's reconstruction error crosses the critical threshold.
2. **Zoom Out for Context:** Never evaluate a spiked day in isolation. If you discover a primary indicator (e.g., Wikileaks Access Count or Keylogger Download Count) on Day 1, immediately check the surrounding timeline. Look for secondary indicators (e.g., after-hours activity, job site browsing, or unauthorized logins) on previous or subsequent days to map the escalation.
3. **Consult the LLM:** Use the Gemini chatbot to clarify complex data tables in Design A or to translate the SHAP feature attributions in Design B.
4. **Determine the Scenario:** A single indicator is just an anomaly; multiple indicators connected across time confirm a malicious scenario. Once you have gathered multi-day evidence, categorize the anomalous action type and submit your final case decision in the case assessment panel.